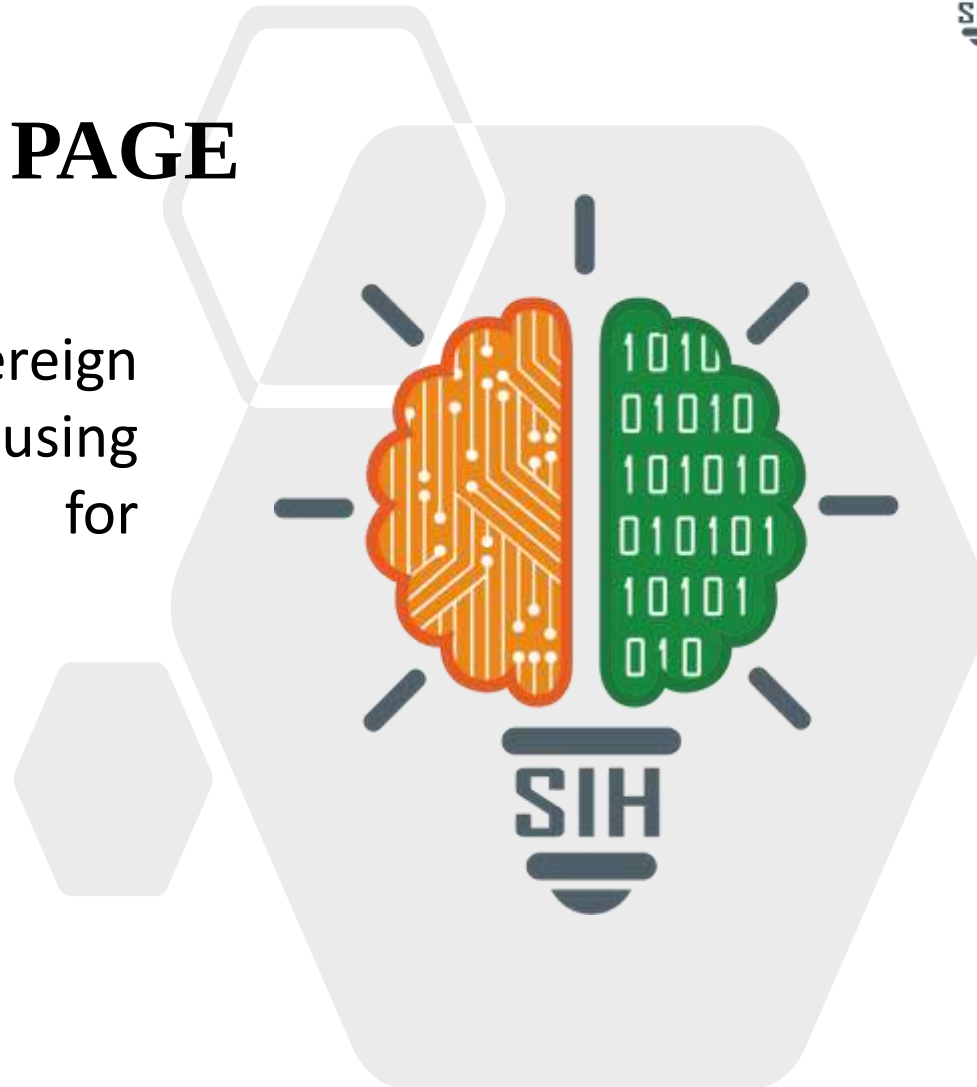


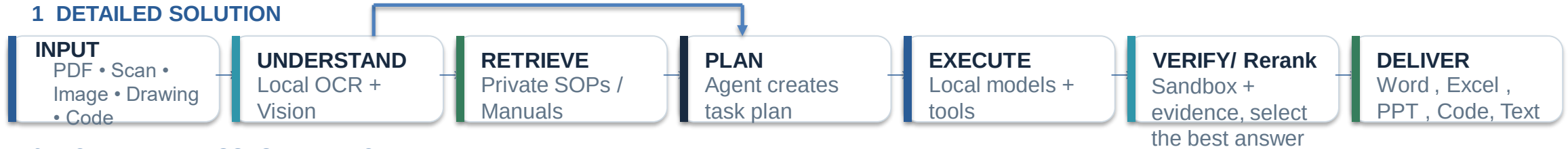
TITLE PAGE

- **Problem Statement ID** – SIH26117
- **Problem Statement Title**- “Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work”
- **Theme** - Smart Automation
- **PS Category**- Software
- **Team ID**-
- **Team Name** – TEAM ADRESTIA

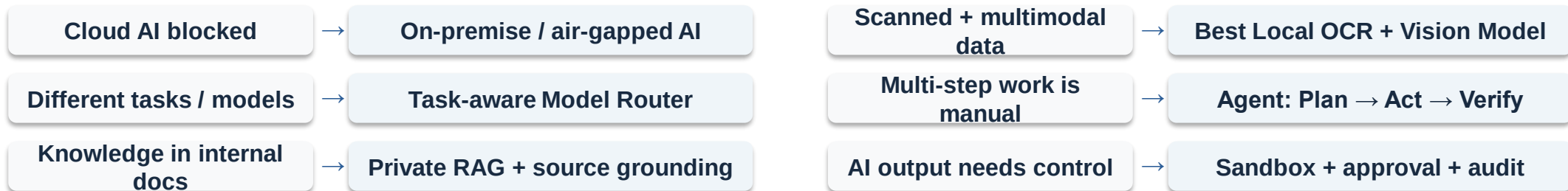


From Confidential Data → Verified Action

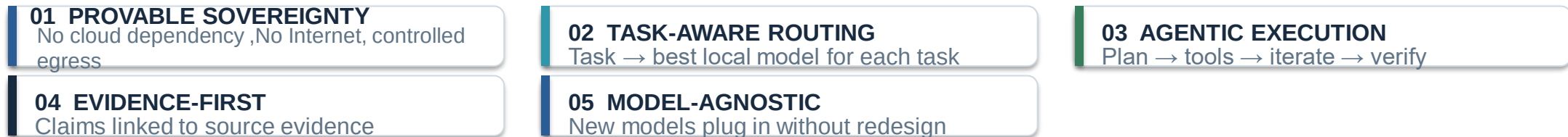
1 DETAILED SOLUTION



2 HOW IT ADDRESSES THE PROBLEM



3 INNOVATION & UNIQUENESS



CONFIDENTIAL DATA → LOCAL AI → VERIFIED WORK → REAL DELIVERABLE

USER REQUEST (INPUT)

(PDF , Image , Engg. Blueprints , Text , Code)



Policy / Security Gate
(check if internet is closed
And perform other security checks)



AGENT ORCHESTRATOR
(Plan → Act → Verify)



MODEL ROUTER



**OCR AI
MODEL**



**CODING AI
MODEL**



**REASONING AI
MODEL**

SANDBOX VERIFICATION



PRIVATE RAG



SOPs Records



Manuals Docs

OUTPUT ANSWER WITH SOURCE DOCUMENT

TEAM
ADRESTIA

FEASIBILITY AND VIABILITY



1. DEPLOY ON EXISTING INFRA

GPU workstation / server
Containerized local services
Air-gapped or restricted intranet

2. MODULAR MODEL LAYER

LLM / VLM / OCR plugins
Task-aware routing
New model can add later without redesigning the whole system

3. CONTROLLED EXECUTION

RBAC + approval gates
Sandboxed tools
Full audit trail

SAFETY – CONTROLLED EXECUTION

- 🛡️ RBAC
- 🔒 No Internet
- ☐ Sandboxed Code
- ✓ Human Approval Gates
- 📁 Audit Logs
- 📌 Evidence-linked Output

FEASIBILITY PROOF

Server

Open
Source AI
Models and
Stack

AGENT + RAG
+ TOOLS

Verified
OUTPUT

WHY VIABLE

OPEN-WEIGHT

No vendor
lock-in

LOCAL
KNOWLEDGE
Private RAG

SAFE TOOLING
Sandboxed
execution

MEASURABLE
Accuracy • latency
• egress

FROM CONFIDENTIAL DATA TO DECISION-READY OUTPUT

TODAY — FRAGMENTED WORK

- Manual review of reports
- Multiple disconnected tools
- Cloud AI blocked by policy
- Answers are hard to audit



WITH OUR WORKBENCH

- One sovereign workspace
- Agent plans + executes + verifies
- Local model selected per task
- Evidence + audit trail + deliverable

TEAM
ADRESTIA

IMPACT AND BENEFITS



SOVEREIGN AI

SECURITY

PRODUCTIVITY

QUALITY

Confidential
data stays
inside



Faster work
automation



Evidence-based
outputs



100%

DATA PRIVACY

85%

DRAFTING TIME SAVED

\$0

CLOUD RECURRING COST

0 KB/s

OUTBOUND LEAKAGE

Impact Industries :-

INDUSTRY

Inspection reports → decisions →
approval documents

PSUs / GOVERNMENT

Sensitive documents processed
without cloud exposure

ENGINEERING

Blueprints, calculations and technical
reports in one workflow

IT

Code generation + sandbox testing +
verification

Your
Team
Name

RESEARCH AND REFERENCES



RESEARCH BASIS + KEY REFERENCES

OPEN-WEIGHT AI

Qwen • Llama • Mistral • Gemma

DOCUMENT INTELLIGENCE

PaddleOCR • OCR + VLM pipelines

RAG / KNOWLEDGE

FAISS • Qdrant • Milvus

AGENTIC ORCHESTRATION

LangChain • LlamaIndex • Haystack

SELECTED REFERENCES

1. Smart India Hackathon 2026 — Problem Statement SIH26117
2. Qwen • Meta Llama • Mistral • Gemma — official model documentation
3. PaddleOCR — official documentation / repository
4. FAISS — Meta AI Research / official repository
5. Qdrant — official documentation
6. LangChain • LlamaIndex • Haystack — official documentation
7. NIST — AI Risk Management Framework (AI RMF)

DESIGN PRINCIPLE → open + locally deployable components can evolve without redesigning the core.

Our architecture combines proven open technologies into one sovereign, multimodal and agentic workflow for confidential industrial work.